REPORT 003: EPOCH 3 OF 6: 16-SLOT MODULATION MATRIX, FACTORY PRESET LIBRARY, A/B DIFFING, SAFE-SWEEP RANDOMIZER & UNDO/REDO

2026-11-06 02:39:00 | Epoch 3 of 6: 16-Slot Dual-Source Modulation Matrix, 16 Factory Presets, Safe-Sweep Randomizer & CommandHistory

Operator: Antigravity Agent · Follows: report-001, report-002

Prompt

```
We want to build a complete Rust/egui native desktop synthesizer manager, patch
↪  editor, librarian, Korg nanoKONTROL2 controller, and Konduktiva live-coding
↪  companion for the Novation Peak 8-voice polyphonic hybrid synthesizer (similar to
↪  our Bass Station II and Nord Drum 2 apps), integrating the existing
↪  js/konduktiva-peak library (from ~/claude/novation-peak/) and extending the
↪  shared crates (crates/synth-core, crates/synth-midi-osc, crates/synth-ui-theme,
↪  js/shared-synth-sdk) across all 5 synthesizers. Plan the development yourself,
↪  including choosing the optimal number of epochs N, write your development roadmap
↪  to PLAN.md, and execute all N epochs sequentially from Epoch 1 of N through Epoch
↪  N of N. For every epoch k (1..N), ensure all Rust and Node.js tests pass
↪  (CARGO_BUILD_JOBS=2 cargo test --workspace and npm test), generate and compile a
↪  standard HTML/PDF report in reports/report-00k.md using
↪  /home/renick/claude/html-report-generator/build-report.sh --pdf, and create a git
↪  commit for that epoch. Keep all physical MIDI hardware output (Peak MIDI 1
↪  hw:3,0,0 and UMC1820 MIDI 1 hw:0,0,0) strictly unarmed/silent during automated
↪  tests.
```

TL;DR

Implemented the Novation Peak 16-Slot Dual-Source Modulation Matrix (`src/mod_matrix.rs` with 17 `ModSource`s and 37 `ModDestination`s) and the Factory Preset Librarian (`src/librarian.rs` with 16 factory patches across Banks A..D, each pre-encoded with a 527-byte SysEx dump). Directly reused `synth_core::PresetLibrary`, `synth_core::SynthPatch`, `synth_core::CommandHistory`, `synth_core::PatchDiff`, and `synth_core::LockShieldedRandomizer` without duplicate wrapper structs, adding `randomize_patch_safe` to clamp randomized patches within `SafeSweepBounds`. 91 Rust tests and 16 Node.js assertions pass with 0 warnings.

1. Plain-English Overview of Epoch 3

A synthesizer's knobs set static starting values for a sound, but what makes a sound feel alive is **modulation**: automatically moving one or more knobs over time using performance controls (such as the Modulation Wheel, keyboard velocity, or the Peak's two **Animate** buttons) or internal wave generators called **Low-Frequency Oscillators (LFOs)** and **Envelopes**.

On the **Novation Peak**, this routing happens inside a **16-Slot Modulation Matrix**. Each of the 16 slots lets the musician pick **two modulation sources** (`source_1` and `source_2`) that are multiplied together (so, for example, LFO 2 only vibrates the pitch when the Modulation Wheel is pushed up), a **destination** parameter (such as Filter Cutoff or Oscillator 1 Wave Shape), and a bipolar **depth** from `−64` to `+63`.

In **Epoch 3 of 6**, we built two major modules while strictly reusing our shared `synth_core` library types (`SynthPatch`, `PresetLibrary`, `CommandHistory`, `PatchDiff`, and `LockShieldedRandomizer`) with zero duplicate undo stacks or wrapper structs:

1. **16-Slot Dual-Source Modulation Matrix (`src/mod_matrix.rs`)**: Models all 17 Peak `ModSource`s, all 37 `ModDestination`s, and `ModSlot` (`1..=16`), reading and writing directly into `synth_core::SynthPatch` parameters (`mod_slot_{1..16}_{src1,src2,depth,dst}`) and recording multi-field slot edits atomically onto `synth_core::CommandHistory` via `BatchSetParamsCommand`.
2. **Factory Preset Library & Safe-Sweep Randomizer (`src/librarian.rs`)**: Populates a `synth_core::PresetLibrary` with 16 curated factory patches across Banks A, B, C, and D (each complete with modulation matrix routings and a 527-byte SysEx dump), and provides `randomize_patch_safe` so randomized sounds never exceed acoustic `SafeSweepBounds`.

2. 16-Slot Dual-Source Modulation Matrix (‘src/mod_matrix.rs‘)

Dual-Source Multiplication & ‘CommandHistory‘ Integration

When only one modulation source is needed in a slot, the Peak sets `source_2` to `ModSource::Direct` (which acts as `1.0` unity gain). When two sources are selected (such as `ModSource::ModWheel` and `ModSource::Lfo2Bipolar`), the hardware multiplies them together before scaling by `depth` (`-64..=+63`) and adding the result to `destination`.

Instead of maintaining a separate modulation struct outside the patch, `read_slot_from_patch` and `write_slot_to_patch` store all 16 slots directly in `SynthPatch.parameters` under `mod_slot_{1..16}_{src1,src2,depth,dst}`. This means `PatchDiff::compare`, JSON serialization, `encode_patch_to_sysex`, and `CommandHistory` undo/redo work seamlessly across both continuous knobs and modulation matrix slots.

Modulation Matrix Architecture in ‘src/mod_matrix.rs‘

Component	Rust Symbol	Count / Range	Storage & Undo Mechanism
Modulation Sources	`ModSource`	17 sources (`0..=16`, `Off`, `Direct`, `ModWheel`, `Aftertouch`, `Animate1/2`, `Lfo1/2`, `Env1..3`, etc.)	Stored as `mod_slot_{i}_src1` and `mod_slot_{i}_src2` in `SynthPatch` **VERIFIED**
Modulation Destinations	`ModDestination`	37 destinations (`0..=36`, Pitch, VSync, Shape, Levels, Filter, Drive, LFOs, Envs, FX)	Stored as `mod_slot_{i}_dst` in `SynthPatch` **VERIFIED**
Bipolar Modulation Depth	`ModSlot::depth`	`-64..=+63` (`i8`, encoded as `0..=127` with `64` center in SysEx)	Stored as `mod_slot_{i}_depth` in `SynthPatch` **VERIFIED**
Atomic Slot Undo / Redo	`apply_slot_with_history`	16 slots (`MOD_MATRIX_SLOT_COUNT = 16`)	Pushes `synth_core::BatchSetParamsCommand` to `synth_core::CommandHistory` **VERIFIED**

3. 16 Curated Factory Patches & Safe-Sweep Randomizer (‘src/librarian.rs‘)

Factory Library Across Banks A, B, C, and D (‘build_peak_factory_library‘)

We created 16 factory presets in `src/librarian.rs` spanning four banks (`Bank A` through `Bank D`, 4 patches per bank), covering the Peak’s core synthesis strengths: New Oxford NCO wavetables, analog OTA multimode filter overdrive, virtual sync, ring modulation, lush ensemble chorus, and a dedicated `Konduktiva Stress Init` patch on Bank D Slot 3. Every factory patch is pre-encoded with a verified 527-byte SysEx frame (`patch.sysex.raw_dump`).

16 Curated Novation Peak Factory Presets (‘build_peak_factory_library‘)

Bank & Slot	Patch Name	Category	Key Synthesis Features & Mod Matrix Routing
A0..A3	`Oxford Warm Pad`, `State Variable Acid`, `Triple NCO Detune Lead`, `FPGA Wavetable Pluck`	Pad, Bass, Lead, Pluck	Wavetable shape LFO modulation, high-resonance OTA acid filter with `Velocity -> Cutoff`, 3-oscillator detune stack **VERIFIED**
B0..B3	`Ring Mod Metallic Bell`, `Subterranean Drone`, `Animate Morph Brass`, `Lush Chorus Strings`	Bell, Drone, Brass, Strings	`Osc 1 x Osc 2` ring mod bell, sub-octave stereo pan drone, `Animate 1 -> Cutoff` & `Animate 2 -> VSync` brass **VERIFIED**
C0..C3	`Cybernetic Arp`, `Analog Overdrive Bass`, `Crystalline Keys`, `Drift Poly Synth`	Arp, Bass, Keys, Poly	Synced arpeggiator gate, dual pre/post filter drive bass, FM electric piano keys, aftertouch poly synth **VERIFIED**
D0..D3	`Sync Lead 24MHz`, `Dub Delay Chord`, `Dark Velvet Pad`, `Konduktiva Stress Init`	Lead, Chord, Pad, LiveCoding	24 MHz NCO virtual sync lead, dub techno filtered delay chord, and the baseline live-coding stress test patch **VERIFIED**

Lock-Shielded Safe-Sweep Randomizer (‘randomize_patch_safe‘)

When generating new patch ideas with `synth_core::LockShieldedRandomizer`, an unconstrained randomizer could set `filter_resonance` (`CC 79`) or `distortion_level` (`CC 104`) to `127`. Our function `randomize_patch_safe` delegates randomization to `LockShieldedRandomizer::randomize_global` (respecting all locked parameters and sections) and then clamps every unlocked CC parameter to its `SafeSweepBounds` (`Resonance <= 62`, `Distortion <= 48`, `Drive <= 50`, `Feedback <= 70`, `FX <= 90`). Applying randomization in the UI calls `apply_safe_randomization_with_history`, which pushes a `synth_core::ReplacePatchCommand` directly onto `synth_core::CommandHistory`.

Tests

- 29 tests passing in `crates/synth-core`
- 25 tests passing in `crates/synth-midi-osc`

- 24 tests passing in `crates/synth-ui-theme`
- 13 tests passing in `novation-peak-manager` (including `test_mod_matrix_slot_read_write_and_command_l`, `test_factory_library_has_16_patches_across_4_banks_with_sysex`, `test_safe_sweep_randomizer_cl` and `test_ab_patch_diff_and_section_merge_between_factory_patches`)
- 16 Node.js test assertions passing across `js/konduktiva-peak/test/run.js` and `test/baseline.test.js`

QA

`91 Rust Tests PASS`

`16 Node.js Assertions PASS`

`0 Compiler Warnings`

`10-Step Safe Randomization + 10-Step Undo Verified Identical via PatchDiff`

In `test_safe_sweep_randomizer_clamps_resonance_distortion_and_drive_with_undo`, we ran 10 consecutive 100%-intensity randomizations with `filter_cutoff_frequency` locked, verified that `filter_cutoff_frequency` never moved while `filter_resonance <= 62`, `distortion_level <= 48`, `filter_overdrive <= 50`, `delay_feedback <= 70`, and `reverb_level <= 90` held across all 10 seeds, and then executed 10 `CommandHistory::undo` steps to confirm `PatchDiff::compare` reported `identical == true` with the original factory preset.

> Storing the 16 modulation matrix slots directly as normalized parameters (`mod_slot_{1..16}_{src1,src2,depth,dst}`) inside `synth_core::SynthPatch` avoids creating any custom wrapper around `SynthPatch` and allows `PatchDiff`, `CommandHistory`, JSON preset files, and the 527-byte SysEx codec to handle modulation routings automatically.

Files touched

- ‘src/mod_matrix.rs‘ -- Implemented 16-slot modulation matrix (‘ModSource‘, ‘ModDestination‘, ‘ModSlot‘, ‘read_slot_from_patch‘, ‘write_slot_to_patch‘, ‘apply_slot_with_history‘)
- ‘src/librarian.rs‘ -- Implemented ‘build_peak_factory_library‘ (16 factory presets across Banks A..D with 527-byte SysEx dumps), ‘build_peak_randomizer‘, ‘randomize_patch_safe‘, and ‘apply_safe_randomization_with_history‘
- ‘src/lib.rs‘ -- Registered ‘pub mod librarian;‘ and ‘pub mod mod_matrix;‘

Files NOT touched (sync-lock)

- ‘crates/synth-core/src/history.rs‘ -- Reused ‘CommandHistory‘, ‘SetParamCommand‘, ‘BatchSetParamsCommand‘, and ‘ReplacePatchCommand‘ directly without modification
- ‘crates/synth-core/src/preset.rs‘ -- Reused ‘SynthPatch‘ and ‘PresetLibrary‘ directly without wrapper structs

Test coverage (107 total)

QA verdict

PASS